

Exploratory Data Analysis: Sample Superstore

Comprehensive Performance, Discount, and Seasonal Insights Report

Prepared For: Nexora Analytics

Dataset: Sample Superstore (9,994 records)

Date: September 2026

1. Executive Summary

This report summarizes the findings from an Exploratory Data Analysis (EDA) conducted on the **Sample Superstore** dataset. The primary objective is to evaluate underlying profit drivers, examine regional and category performances, and analyze the financial impact of current discounting policies.

TOTAL REVENUE

\$2.30M

9,994 Orders

TOTAL PROFIT

\$286.4K

12.4% Net Margin

TOP CATEGORY

Technology

\$145.5K Profit

AVG DISCOUNT

15.6%

Loss @ ≥ 20%

2. Key Findings & Quantitative Insights

A. Profitability & Category Analysis

Profitability varies significantly across the three main product categories. **Technology** and **Office Supplies** represent the principal profit drivers, generating the majority of net revenue. Conversely, **Furniture** suffers from suppressed margins due to high shipping expenses and aggressive discounting on sub-categories such as Tables and Bookcases.

Product Category	Total Sales (\$)	Total Profit (\$)	Profit Margin (%)	Performance Status
Technology	\$836,154.03	\$145,454.95	17.39%	High Performing
Office Supplies	\$719,047.03	\$122,490.80	17.03%	High Performing
Furniture	\$741,999.79	\$18,451.27	2.49%	Underperforming

B. Discount vs. Profit Impact Analysis

Statistical analysis reveals a strong negative correlation ($r = -0.38$) between discount rates and order profitability. While moderate discounts (0% – 10%) preserve profit margins, discount levels equal to or exceeding **20% (0.20)** consistently result in negative average profits per transaction.

Discount Tier	Order Count	Average Profit (\$)	Total Profit (\$)	Status
0% (No Discount)	4,798	+\$66.90	+\$320,987.00	Highly Profitable
10% - 15%	582	+\$15.20	+\$8,846.40	Profitable
20% Threshold	3,657	-\$5.10	-\$18,650.70	Loss Making

Discount Tier	Order Count	Average Profit (\$)	Total Profit (\$)	Status
30% - 80%	957	-\$86.40	-\$82,684.80	Severe Loss

C. Seasonal Sales Patterns

- **Q4 Peak Surge:** Sales volume peaks heavily in September, November, and December due to corporate year-end budget utilization and retail holiday demand.
- **Q1 Annual Trough:** January and February record the lowest total revenue across all four observed years.

3. Strategic Business Recommendations

1. Enforce a 15% Discount Cap

High Priority

Implement system-level checks in the order processing pipeline to cap standard promotional discounts at 15%. Any custom pricing above 15% must require explicit approval from a regional manager.

2. Restructure Furniture & Sub-Category Pricing

High Priority

Re-evaluate supplier pricing and logistics models for Tables and Bookcases. Discontinue standalone high-discount offers on low-margin furniture lines.

3. Align Inventory with Q4 Demand Seasonality

Medium Priority

Scale up logistics capacity and stock levels starting in August to capture peak Q4 demand. Introduce early Q1 clearance programs to smooth out the post-holiday sales dip.

4. Shift Regional Strategy to Margin-Focused Metrics

Medium Priority

Transition sales incentives in underperforming regions (e.g., Central Region) from gross volume targets to net profitability metrics to curb excessive discounting practices.

5. Drive Automated Cross-Selling

Growth Opportunity

Bundle high-margin Office Supplies (Paper, Envelopes) with high-value Technology hardware purchases during checkout to raise average order value without discounting.